THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM EXAMINATION

FORM ONE MID-TERM EXAMINATION
043 MATHEMATICS
MARKING SCHEME AUGUST 2026

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1. (a) Prime numbers are: 2, 3, 5, 7.....01 mark.
The number formed in ascending order is 2357.....01 mark.
Square numbers are 1, 4 and 9.....01 mark.
The number formed in descending order is 941.....01 mark.
$M - N = 2357 - 941 = 1\,416$.....01mark.
(b) $= 125 - 3(-2) + 16$.....01 mark.
$= 125 + 6 + 16$.....01 mark.
$= 125 + 22$.....01 mark.
$= 147$.....02 marks.

2. (a) Total ratios$= 6 + 8 + 10 = 24$ parts.....01 mark.
Difference between the largest and smallest $= \left(\frac{10-6}{24}\right) \times 288\ cm$.....01 mark.
$= \frac{4}{24} \times 288\ cm$.....01 mark.
$= 48\ cm$ ....02 marks.

(b) David : Amina $= 7:3$
David : Nuru$= 4:5$
Let the oranges received by Nuru be x
Therefore $x: x + 21 = 4:5$.....01 mark.
$\frac{x}{x+21} = \frac{4}{5}$ .....01 mark.

$5x = 4x + 8$

$5x - 4x = 84$

$x = 84$.....01 mark.

Let the number of oranges received by Nuru be y
$84 : y = 7:3$
$\frac{84}{y} = \frac{7}{3}$ .....01 mark.
$y = \frac{3 \times 84}{7} = 36$
Therefore Amina received 36 oranges.....01 mark.

3. (a)
i. 8 917 +201 392 +500 099 + 1 983=712 391.....02 marks.
$712\,391 \approx 712\,000$.....02 marks.
ii. 0.5021 +21. 1108 +0.00472+3. 079 $= 24.69662$.....02marks.
$24.69662 \approx 24.7$.....02 marks.

(b)

i. $5.1825 \times 20 = 103.65 \approx 103.6$.....01 mark.

ii. $149.001 \div 8.99 = 16.574\ldots \approx 20$.....01 mark.

iii. $\frac{1}{80} = 0.0125 \approx 0.01$.....01 mark.

iv. $0.83 \times 0.737 = 0.61171 \approx 1$.....01 mark.

4. (a) let the cost of a pair of trouser be x

Let the cost of a pair of trouser be y

$$\left.\begin{matrix} 3x + 4y = 12000 \\ 7x + 12y = 32000 \end{matrix}\right\}$$ 01mark

$$\begin{cases} 3x + y = 12000 \\ 7x + 12y = 32000 \end{cases}$$

$$9x + 12y = 36000$$
$$7x + 12y = 32000$$

$$- \left| \begin{matrix} 9x + 12y = 36000 \\ 7x + 12y = 32000 \end{matrix} \right.$$

$$2x = 4000$$

$$\frac{2x}{2} = \frac{4000}{2}$$

$x = 2\,000$ .....01 mark.

Substitute 2000 for x in $3x + 4y = 12000$

$3 \times 2000 + 4y = 12000$

$6000 + 4y = 12000$

$4y = 12000 - 6000$

$4y = 6000$ .....01 mark.

$y = \frac{6000}{4}$

$y = 1500$ .....01 mark.

∴One pair of trouser costs Tsh. 2000 while a skirt costs Tsh 1500 .....01 mark.

(b) $\overline{AB} = \overline{BC}$ (Isoscles triangle)

$$5x - 3y = 3x + 2y + 1$$

$$5x - 3x - 3y - 2y = 1$$

$2x - 5y = 1$.....equation 1.....01 mark.

Perimeter= $\overline{AB} + \overline{BC} + \overline{AC}$

$$29 = 5x - 3y + 3x + 2y + 1 + x + 2y$$

$$29 - 1 = 9x + y$$

$28 = 9x + y$.....equation 2.....01 mark.

$$\begin{cases} 2x - 5y = 1 \\ 9x + y = 28 \end{cases}$$

By calculator $x = 3 \; and \; y = 1$.....02 marks.

$\overline{CA} = x + 2y = 3 + 2 \times 1 = 5 \; cm$..... .....01 mark.

5. (a)

i. $2.5555\ldots = 2.\dot{5}$ .....$00\frac{1}{2}$ mark.

ii. $3.040404\ldots = 3.\dot{0}\dot{4}$.....$00\frac{1}{2}$ mark.

iii. $\frac{1}{12} = 0.083333\ldots = 0.08\dot{3}$.....01 mark.

iv. $2.141414\ldots + 0.83333\ldots = 2.9\dot{7}\dot{4}$.....01 mark.

(b) The largest decimal is $3.\dot{0}\dot{4}$.....01 mark.

$let \; y = 3.\dot{0}\dot{4}$

$y = 3.\dot{0}\dot{4}$

$100y = 3.\dot{0}\dot{4} \times 100 = 304.\dot{0}\dot{4}$ .....01 mark.

$$-\begin{vmatrix} 100y = 304.\dot{0}\dot{4} \\ y = 3.\dot{0}\dot{4} \end{vmatrix}$$

$99y = 301$ ....01 mark.

$\frac{99y}{99} = \frac{301}{99}$

$x = \frac{301}{99}$.....01 mark.

(c) **Mathematics in Geography for Personal Income (3 Marks)**

(i) **Map Making and Cartography:** Using scale drawings, ratios, and coordinates to draw accurate local maps for businesses or tourists. *Example:* A student calculates accurate scale models to design and sell custom hiking or tourist maps of a local national park.

(ii) **Weather and Climate Data Analysis:** Using statistics, percentages, and graphs to calculate average rainfall and temperatures for local farmers. *Example:* Charging a fee to help commercial farmers look at temperature graphs so they know exactly which weeks are best to plant crops.

(iii) **Surveying and Land Measurement:** Using geometry, shapes, and basic trigonometry to measure land boundaries and calculate the exact area of building plots. *Example:* Working as an assistant surveyor to calculate the square metres of a plot of land before it is sold.

6. (a) $= \frac{-17}{2} + \frac{1}{10} - \frac{23}{7}$

$=- 8\frac{1}{2} + \frac{1}{10} - 3\frac{2}{7}$

$= (-8 - 3) + \left(\frac{-1}{2} + \frac{1}{10} - \frac{2}{7}\right)$.....01 mark.

$=- 11 + \left(\frac{-1}{2} - \frac{2}{7}\right) + \frac{1}{10}$

$=-11+\left(\frac{-7-4}{14}\right)+\frac{1}{10}$

$=-11+\left(\frac{-11}{14}\right)+\frac{1}{10}$.....01 mark.

$=-11+\left(\frac{-55+7}{70}\right)$

$=-11-\frac{48}{70}$

$=-11-\frac{24}{34}$

$=-11\frac{24}{35}$ .....01 mark.

$\left|-11\frac{24}{35}\right|=11\frac{24}{35}$.....02 marks.

(b)$\pm\left(\frac{-2}{3}x-4\right)<2$

$\frac{-2}{3}x-4+4<2+4$

$\frac{-2}{3}x<6$.....$00\frac{1}{2}$ mark.

$\frac{-2}{3}x\times\frac{3}{-2}>6\times\frac{3}{-2}$

$x>-9$ .....01 mark.

or

$\frac{2}{3}x+4<2$

$\frac{2}{3}x+4-4<2-4$

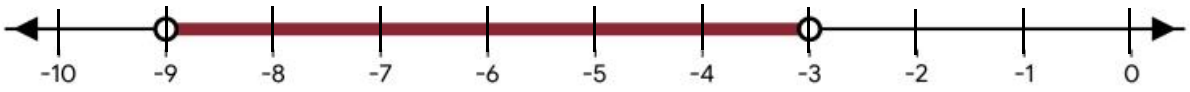
$\frac{2}{3}x<-2$.....$00\frac{1}{2}$ mark.

$\frac{2}{3}x\times\frac{3}{2}<-2\times\frac{3}{2}$

$x<-3$.....01 mark.

.....2 marks

$-9<x<-3$



7. (a) $\frac{1}{\frac{8}{7}}=1\div\frac{8}{7}=1\times\frac{7}{8}=\frac{7}{8}=0.875$.....01 mark.

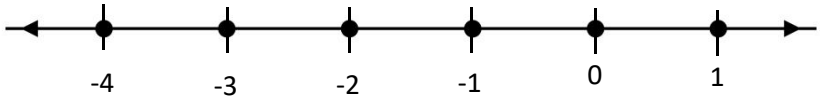
$40\%=\frac{40}{100}=0.4$ .....01 mark.

$0.5\dot{2}=0.5\dot{2}$.....01 mark.

Therefore $40\%, 0.5\dot{2}$, and $\frac{1}{\frac{8}{7}}$ are in ascending order.....02 *marks*.

(b) i. Integers are $-4, -$ Graph here$3, -2, -1, 0$ .....01 mark.

.....2 marks



ii. .....2 marks



8. (a) i. $\begin{matrix}1\\2\end{matrix}\left|\begin{matrix}5x+2y=6\\3x-y=8\end{matrix}\right.$

$5x+2y=6$

$6x-2y=16$ .....0 mark.

$+\left|\begin{matrix}6x-2y=16\\5x+y=6\end{matrix}\right.$ .....01 mark.

$11x=22$

$\frac{11x}{11}=\frac{22}{11}$

$x=2$ .....01 mark.

Substitute 2 for $x\ in\ 3x-y=8$

$3\times 2-y=8$

$6-y=8$.....01 mark.

$6-8=y$

$-2=y$ .....01 mark.

(b) $\begin{matrix}5x+2y=6\\3x-y=8\end{matrix}$

From $3x-y=8$

$y=3x-8$.....01 mark.

Substitute $3x-8\ for\ y\ in 5x+2y=8$

$5x+2(3x-8)=6$.....01 mark.

$5x+6x-16=6$

$11x=22$

$\frac{11x}{11}=\frac{22}{11}$

$x=2$.....01 mark.

Substitute 2 for $x\ in\ y=3x-8$

$y=3\times 2-8$.....01 mark.

$=6-8$

$=-2$.....01 mark.

9. (a) $x=6y$ .....01 mark.

$4y+4=6y$

$4=6y-4y$

$4=2y$.....01 mark.

$\frac{2y}{2}=\frac{4}{2}$

$y=2$.....01 mark.

$x = 6y = 6 \times 2 = 12$.....02 marks.

(b) $= 2\,618\,954 + 124 \times 68 - 20243$.....01 mark.
$= 2\,618\,954 + 8\,432 - 20243$.....01 mark.
$= 2\,627\,954 - 20243$.....01 mark.
$= 2607143$.....01 mark.
$2607143 \approx 2\,610\,000$.....01 mark.

10. (a)**Time Management** .....05 marks.

- **Planning Daily Schedule:** We use addition and subtraction to divide our 24-hour day into different blocks for studying, sleeping, and playing. *Example:* Adding 2 hours of homework time to 4 hours of school activities to see how much of the evening is left.
- **Meeting Deadlines:** *Explanation:* We calculate the number of days or hours remaining before an exam or project is due so we can plan our preparation. *Example:* Subtracting today's date from the exam date to find out exactly how many days are left to revise.
- **Travel and Punctuality:** We calculate speed, distance, and time to know exactly what time we need to leave home to arrive at school or a meeting punctually. *Example:* Knowing that walking takes 20 minutes, so subtracting 20 minutes from the school start time to decide the departure time.
- **Prioritising Tasks (Using Percentages/Ratios):** We use fractions or percentages to decide how much total effort or time to give to a difficult subject compared to an easy one. *Example:* Spending 60% of study time on Mathematics and 40% on Geography.
- **Measuring Efficiency and Speed:** We count or time how long it takes to complete a single task to estimate how long a larger amount of work will take. *Example:* If it takes 5 minutes to read 1 page of a textbook, calculating that reading 10 pages will take 50 minutes.

(b) **Budgeting** .....05 marks.

- **Balancing Income and Expenses:** We use basic addition to find total income and total expenses, then use subtraction to make sure we do not spend more money than we have. *Example:* Subtracting total shopping expenses from pocket money to ensure the balance is not below zero.
- **Allocating Money to Categories:** We use percentages or fractions to divide a total amount of money into specific groups like food, transport, and school supplies. *Example:* Setting aside 1/2 monthly pocket money for school snacks and 1/4 for bus fare.
- **Calculating and Planning Savings:** We use subtraction to find out what money is left over after buying needs, which can be saved for the future.
  - *Example:* Planning to save 1,000 shillings every week from pocket money to buy a new school bag after two months.
- **Finding Best Prices (Comparison Shopping):** We use division to find the unit price of items to see which shop offers the cheaper option or better discount. *Example:* Comparing if a pack of 3 pens for 1,500 shillings is cheaper per pen than a single pen sold for 600 shillings.

- **Preventing Financial Debt:** We use simple mathematical estimation and inequalities (greater than/less than symbols) to make sure our total wants do not cross our financial limit.*Example:* Keeping track of total costs while picking items in a shop so the final bill stays less than (<) the cash in hand.